Lecture Scribe L15-17

lecture1.pdf
PDF

L15_L17.pdf
PDF

EXAMPLE LECTURE SCRIBE (STRUCTURE REFERENCE):
lecture1.pdf

INSTRUCTION:
Using the exact same structure and formatting as the example scribe above, generate a comprehensive lecture scribe for L15-17 based on the provided lecture materials.

PURPOSE:
This scribe will serve as a student's primary exam reference document. A student studying only this document should be able to:
- Recall all definitions, notation, and conventions introduced
- Reconstruct every proof and derivation step-by-step
- Follow all example solutions with clear intermediate calculations
- Understand how concepts build upon each other
- Locate any result or explanation quickly

CONTENT RULES:
- Include ONLY what appears in the provided lecture materials
- Include ALL definitions, theorems, proofs, derivations, and worked examples exactly as presented
- Include EVERY example from the lecture with complete step-by-step solutions
- Do NOT add external explanations, alternative proofs, or supplementary examples
- Do NOT add commentary, study tips, or explanatory notes beyond what's in the lecture
- Do NOT simplify or rephrase proofs—preserve the original level of detail
- Do NOT omit any examples, even if they seem repetitive

STRUCTURE REQUIREMENTS:

Use clear section headings for each major topic. For each component:

Definitions and Notation:
- Precise mathematical statements with symbol definitions
- Domain, range, and any assumptions/conditions

Theorems and Results:
- Clear statement separating hypothesis and conclusion
- Step-by-step proof with each step justified
- References to definitions used

Worked Examples:
- EVERY example from the lecture must appear
- Problem statement followed by step-by-step solution
- ALL intermediate calculations shown—no skipped steps
- Final answer clearly indicated
- Label examples sequentially (Example 1, Example 2, etc.)

Derivations:
- Show each algebraic or logical step
- Include brief justifications for non-obvious steps

Logical Flow:
- "Recall from previous section..." statements where appropriate
- "This leads to..." transitions showing dependencies
- Prerequisite concepts noted before new material

FORMATTING:
- Match the example's heading hierarchy and indentation
- Use consistent notation conventions
- Preserve any visual structuring (boxes, spacing, etc.)

MATERIALS TO PROCESS:
L15-17.pdf

Here is your **Lecture Scribe for L15–17** following the **same structure and style as the reference scribe**, while strictly adhering to the provided slides.

CSE400 — Fundamentals of Probability in Computing

Lecture L15–17 Scribe

Joint Random Variables + Joint Distributions + Examples

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1 Introduction

These notes reproduce the **technical content** of Lectures L15–17. The focus is on joint random variables, their distributions, and associated functions.

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2 Why Study Joint Random Variables?

We are often interested in analyzing **two or more random variables simultaneously**.

2.1 Motivating Examples

Example 1: Smart Transportation

Let:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question: Can speed be high when traffic density is high?

Conclusion:

- Yes / No (depends on system conditions)
- Therefore, X and Y are **correlated random variables**

Applications:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Let:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Observation:

- Throughput depends on SNR
- Both vary due to:
  - Fading
  - Interference
  - Congestion

Question: If SNR is high, is throughput always high?

Answer: No

Reason: Network congestion introduces dependency

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Example 3: Weather Prediction

Let:

- $X =$ Rainfall
- $Y =$ Temperature

Application: Weather forecasting

Example 4: Drone Delivery

Let:

- $X =$ Wind Speed
- $Y =$ Delivery Time

Application: Logistics and UAV systems

Example 5: Rolling Two Dice

Let:

- $X =$ outcome of first die
- $Y =$ outcome of second die

Example conditions:

- $X + Y = 7$
- $X > Y$

This requires a **joint PMF**

Conclusion

In real-world applications, we frequently deal with **multiple random variables simultaneously**, requiring joint distributions.

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3 Discrete Case — Joint PMF

3.1 Joint PMF Representation

For two discrete random variables X and Y, the **joint probability mass function (PMF)** describes:

$$P(X = x, Y = y)$$

3.2 Joint PMF Table Representation

The joint PMF can be represented in a **table format**:

- Rows correspond to values of X
- Columns correspond to values of Y
- Each entry represents $P(X = x, Y = y)$

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4 Worked Example — Rolling Two Dice

Example 1

Experiment: Roll two dice

Let:

- $X =$ value on first die
- $Y =$ value on second die

Step 1: Sample Space

Total outcomes:

$$(1,1),(1,2),\ldots,(6,6)$$

Total = 36 outcomes

Step 2: Joint PMF

Since dice are fair:

$$P(X=x,Y=y)=\frac{1}{36}$$

for all $x,y \in \{1,2,3,4,5,6\}$

Step 3: Table Representation

Each entry in the table:

$$P(X=x,Y=y)=\frac{1}{36}$$

Step 4: Example Conditions

Case 1: $X+Y=7$

Valid pairs:

$$(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)$$

Number of outcomes = 6

$$P(X+Y=7)=\frac{6}{36}=\frac{1}{6}$$

Case 2: $X > Y$

Count all pairs where first die is greater than second:

$$(2,1),(3,1),(3,2),\ldots$$

Total count = 15

$$P(X>Y)=\frac{15}{36}$$

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5 Continuous Case — Joint PDF

5.1 Definition

For continuous random variables X and Y, the **joint probability density function (PDF)** is:

$$f_{X,Y}(x,y)$$

5.2 Interpretation

- Represents density over a region
- Probability is obtained via **integration over a region**

5.3 Geometric Interpretation

- Probability corresponds to **volume under the surface**
- Region in xy-plane defines limits

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6 Worked Example — Continuous Case

Example 2

Given a joint PDF over a region.

Solution Steps

Step 1: Identify Region

Determine limits of x and y

Step 2: Verify PDF

Check:

$$\iint f_{X,Y}(x,y)\,dx\,dy = 1$$

Step 3: Compute Probability

For a given region R:

$$P((X,Y) \in R) = \iint_R f_{X,Y}(x,y)\,dx\,dy$$

Step 4: Evaluate Integral

Perform integration step-by-step over limits

Final Answer:

Obtained after evaluating the double integral

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7 Joint Cumulative Distribution Function (Joint CDF)

7.1 Definition

The **joint CDF** of X and Y is:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

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8 Joint CDF — Continuous Case

8.1 Expression

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, dv\, du$$

8.2 Interpretation

- Represents cumulative probability up to point (x, y)
- Computed using double integration

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9 Joint CDF — Discrete Case

9.1 Expression

$$F_{X,Y}(x, y) = \sum_{u \leq x} \sum_{v \leq y} P(X = u, Y = v)$$

9.2 Interpretation

- Sum of probabilities for all pairs satisfying:

$$u \leq x, \quad v \leq y$$

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10 Properties of the Joint CDF

10.1 Basic Properties

1. $0 \leq F_{X,Y}(x, y) \leq 1$
2. Non-decreasing in both variables:
   - Increasing x or y does not decrease value
3. Limits:
   - As $x \to \infty, y \to \infty$,

$$F_{X,Y}(x, y) \to 1$$

4. Boundary conditions:
   - As $x \to -\infty$ or $y \to -\infty$,

$$F_{X,Y}(x, y) \to 0$$

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11 Summary Flow

- Real-world problems involve **multiple random variables**
- Leads to:
  - Joint PMF (discrete case)
  - Joint PDF (continuous case)
- From these, we define:
  - Joint CDF
- Applications involve:
  - Probability calculations over regions
  - Understanding relationships between variables

End of Lecture Scribe (L15–17)